

Tammer Haddad

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Pending Interim Secret Clearance

EDUCATION

Northeastern University, Boston, MA Sep. 2023 – Present
Khoury College of Computer Sciences Expected Graduation: May 2028
Candidate for B.S. in **Computer Science** and **Economics**, Minor in **Robotics** | **GPA: 3.49**
Relevant Coursework: Artificial Intelligence, Algorithms and Data Structures, Object Oriented Design, Law and Ethics in Cybersecurity, Computer Systems, Discrete Structures, Intro to Business Statistics, Applied Economics
Activities: Open Volleyball MB, Club Spikeball, 11lb Combat Robotics Team Lead 2024

Harvard University Extension: Calculus 3, Linear Algebra (MATH E-21a&b) Aug. 2022 – May 2023
MIT Beaver Works: Serious Games with AI | Top Team Jul. 2022 – Aug. 2022
MIT Edgerton Center: Digital Logic with Arduino & Intro to CAD Jul. 2020

EXPERIENCE

Club Robotics Intern | BJ's Wholesale Club, Marlborough, MA Jun. 2026 – Aug. 2026

- Built the company's **on-shelf availability** pipeline end to end in **Databricks (PySpark, SQL)**, joining **800M+ Simbe** robot shelf-scan rows across **267 clubs** against enterprise inventory and forecast data
- Quantified **7.2M units** never sold since Jan. 2025 — **\$116.5M** at listed retail, ~**\$14M** in lost margin annually — splitting each loss into **replenishment execution** vs. **allocation** failure
- Handed the pipeline to a full-time owner with **advanced Tableau dashboards** now used by other departments

Robotics AI Researcher | Undergraduate Research Fellowships, Boston, MA Jan. 2025 – Present

- AJC Merit Research Scholar** conducting **Human-Robot Interaction** research under Professor Tan Zhi; completed full-time Spring 2025 co-op, continuing part-time through academic year
- Engineered end-to-end **autonomous assistant robot** on **Hello Robot Stretch 3** platform, building complete **ROS 2** architecture from scratch integrating **LLM-powered intent recognition**, cascading **STT-NLP-TTS** pipeline, **facial recognition**, **autonomous navigation**, and **manipulation capabilities**
- Deployed system as lab's **production framework**, now serving as foundation for ongoing robot development
- Conducted **user studies** on human-robot communication and proxemic behaviors to inform system design

Executive Board of Directors | Northeastern Entrepreneurs Club, Boston, MA Jan. 2024 – Feb. 2025

- Led **community of 2000+** as Co-Director of Massachusetts' **#1 entrepreneurship club (#6 globally)**
- Secured **\$3000+** in sponsorships through direct negotiation with industry partners and external organizations
- Organized cross-disciplinary workshops and **startup hackathons** building practical entrepreneurial skills

Data Analyst Intern, Team Lead | Aspire Institute, Remote May 2024

- Analyzed **30,000+ recipient** newsletter campaigns achieving **44% open rates**; mapped click-through across **15+ content channels** and delivered strategic recommendations in **Excel** and **Tableau**

PROJECTS

Self-Hosted AI Infrastructure Server | **Linux, Docker, Python** Jul. 2026 – Present

- Rebuilt a spare laptop into an always-on **Ubuntu/Docker** server: **20+** containerized services and public sites behind a **Cloudflare** tunnel, on **systemd**-scheduled nightly automation
- Wrote a **Python** agent framework with **tiered LLM failover** (hosted API → self-hosted gateway pool → local **GPU** model), so autonomous research, briefing, and captioning agents survive losing any provider
- Engineered for unattended uptime: health monitoring and alerting, container self-healing, **RAPL** power capping with a reactive thermal governor, and nightly offsite backup

Local Photo Search Engine | **Python, PyTorch, SQLite** Jan. 2026

- Built full-stack semantic search system for **20k+ photos** combining **Florence-2** captioning, **CLIP** embeddings for natural language queries, **InsightFace** face clustering, and **perceptual hashing** for deduplication

AI Pandemic Simulation | **Python** Aug. 2022

- Led **5-person team** developing **award-winning** pandemic simulation using **Minimax algorithm** to optimize containment strategies across infection dynamics and resource allocation variables

COMPUTER KNOWLEDGE

Languages: Python, ROS 2, C++, C, Java, SQL, Kotlin, HTML, CSS, JavaScript, Swift
Technologies: PyTorch, Databricks, PySpark, Tableau, Docker, Linux, Git, Gym, Blender, MS Office, Adobe Suite, Onshape